

Impact of phase assemblage and grain morphology on retained austenite stability

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Summer Research Project

Anupam Saha & Timken Steel Co.

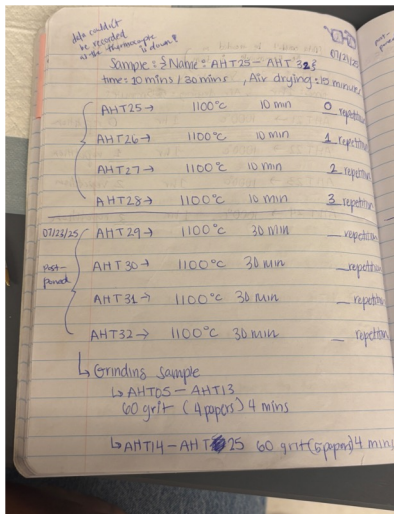
Worked in Dr. Frame's Lab on the Timken Steel Project

Goal: Stabilize retained austenite by increasing carbon content

Gained hands-on experience in:

- Steel processing and microstructure
- Metallurgical techniques (grinding, polishing)

- Investigated M50 steel microstructure at:
- **900°C**
- **1000°C**
- **1100°C**
- Observed thermal effects on grain structure
- Analyzed microstructural differences between temperatures



Sample log & Temperature records



4 metal samples heat treated & labeled



Sample placed inside new furnace

Key Techniques I learned

Grinding & polishing metal samples

Mastered:

- Grit paper selection
- Scratch detection
- Consistent surface finish

Heat treating samples & using lab safety tools

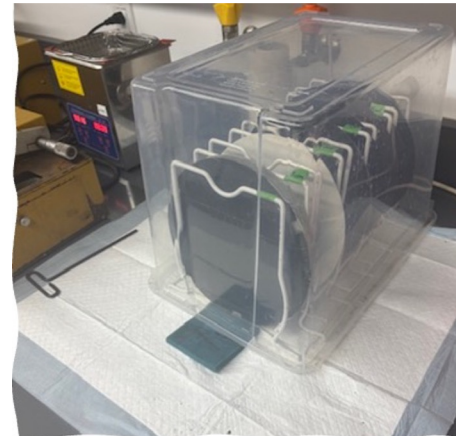
Air quenching vs. oil quenching



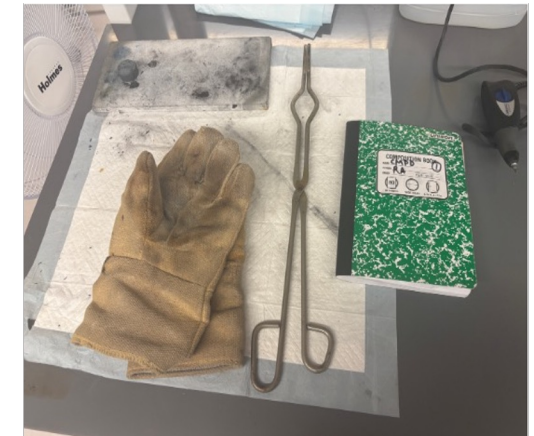
Grinding/Polishing wheel I used



Various Grit sizes available for grinding



Various Polishing wheels

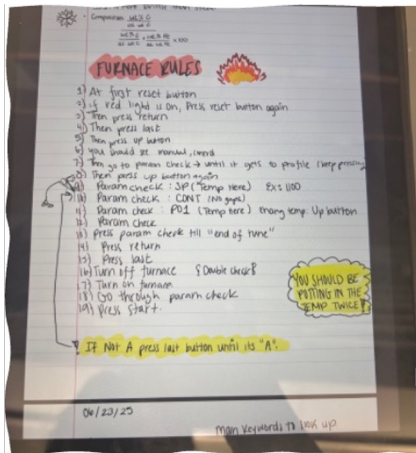


Safety tools required for heat treatment

Heat Treating

- Heat treated at: 900°C, 1000°C, 1100°C
- Switched from oil quenching to air quenching
- Oil quenching caused cracks (rapid cooling)
 - Air quenching reduced cracking risk

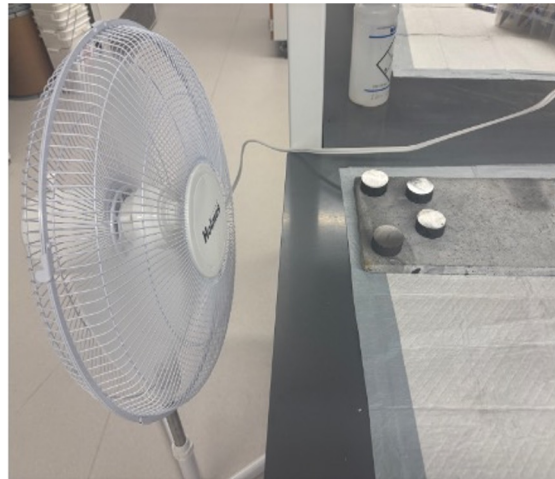
Noticed blue oxide layer at 600°C tempering (surface oxidation)



Rules I made to safely turn on the furnace



Heat treating 4 samples at 1100°C in Furnace



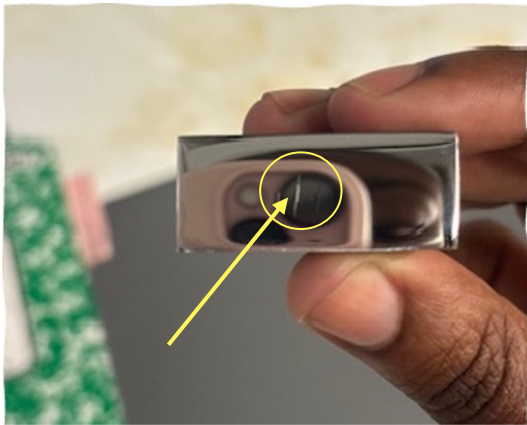
Air quenching process using a fan



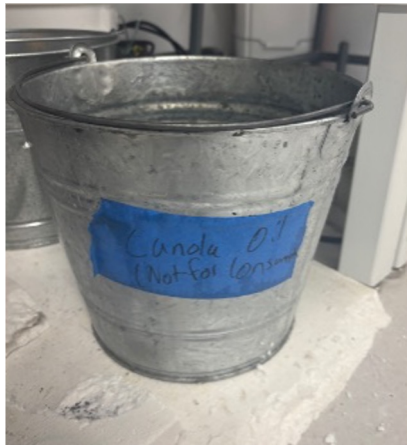
Carving heat treated sample with blue tint

Why We Chose Air Quenching Over Oil Quenching UCONN

- Oil quench (1100°C , 60 mins) caused cracks
- Air quenching = slower cooling $\rightarrow$ fewer cracks
- Found Austenite % was similar in both methods



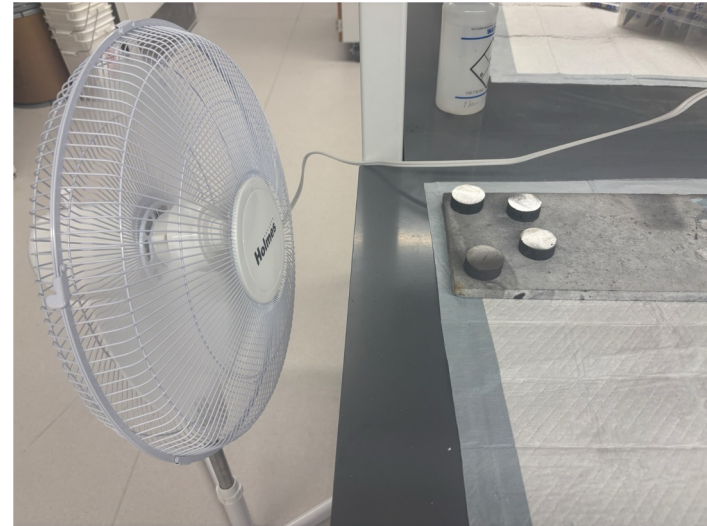
Cracked sample caused by oil quenching at 900°C



Used for initial oil quenching



Quenching medium that caused cracks

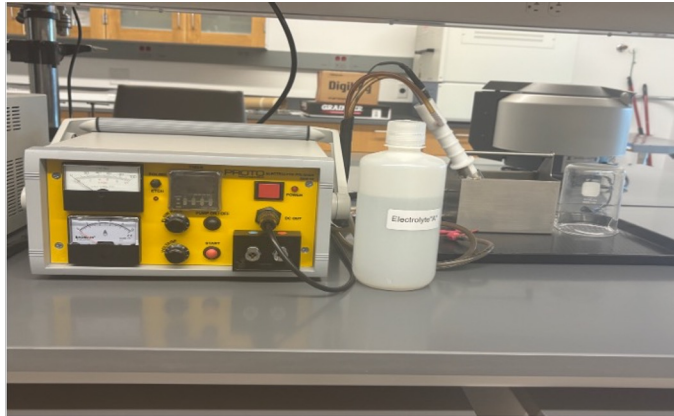


Introduction of Fan setup— had slower & safer cooling rate

Learned about electropolishing (not performed myself)

Purpose:

- Smooth surface post-polishing
- Remove microscopic deformation
- Prep for accurate microscopy



Equipment used for Electropolishing



Grinded Samples ready for electropolishing



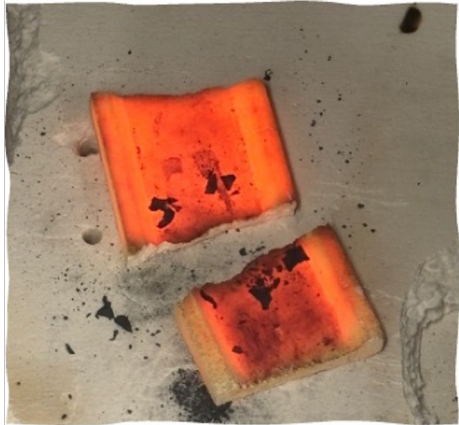
Sample electropolished and ready for microscopy

Personal Struggles

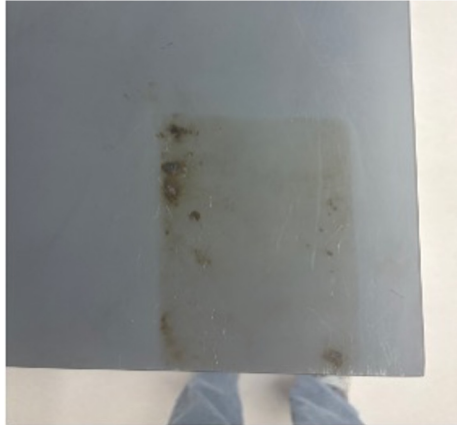
Struggled with polishing in early stages

Common issues:

- Uneven Grinding
- Handling hot objects
- Recognizing proper finish



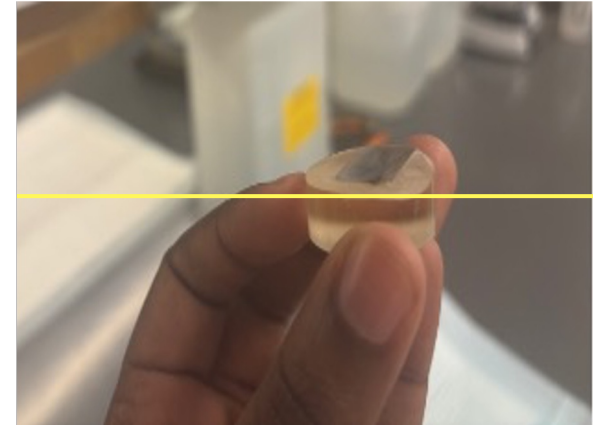
Brick cracking after very last heat treatment– possibly due to repeated cycles of heat treatments.



Burn mark left on table– due to mishandling of hot brick with heated samples.



Early stages of grinding– showcasing uneven surface



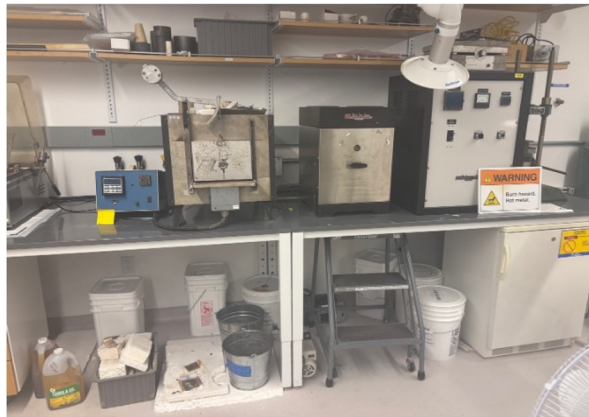
Another example of uneven surface grinded into the sample

Gained confidence with lab tools
Improved micrograph interpretation
Learned:

- Troubleshooting techniques
- Lab discipline and consistency



Tray of over 30+ evenly grinded and heat treated samples



Initial furnace broke down— due to my heat treatment skills I was able to successfully move the bigger furnace without any issues.



One of the samples I grinded and polished with no imperfections or cracks

- Overall, my time in Dr. Frame's lab was challenging but incredibly rewarding. I started with very little lab experience, and by the end, I had completed a full research cycle from preparing samples to analyzing results. I'm proud of how far I came and how much I learned, both technically and personally. This project pushed me to think critically, stay persistent, and believe in my ability to figure things out, even when it wasn't easy. I got hands-on exposure to industry-relevant testing methods and heat treatments, and now I feel much more prepared for future internships or research opportunities.